Phase 2 Modeling Summary

Phase 2: Full-dataset modeling summary

Dataset: all 2,880 rows

Validation: 70/30 stratified holdout + repeated grouped CV on training split

Feature sets compared: embeddings only vs embeddings plus region and Dynamic World context

Current implementation differs from the original proposal: it uses the full dataset and a LightGBM ablation rather than the earlier 720-row subset plus ridge baseline plan.

Best held-out feature set by target:

- S1_VH: Embeddings only with R2=0.942, RMSE=1.182, MAE=0.858, r=0.971

- S1_VV: Embeddings only with R2=0.902, RMSE=1.255, MAE=0.825, r=0.951

- S1_VV_div_VH: Embeddings + region + Dynamic World with R2=0.698, RMSE=1.155, MAE=0.798, r=0.836

Interpretation:

- S1_VV is best with Embeddings only.

- S1_VH is best with Embeddings only.

- VV/VH ratio is best with Embeddings + region + Dynamic World.

- Mean grouped-CV R2 across the selected models ranges from 0.713 to 0.934.

- Important caveat: the outer holdout is stratified random within sampled regions, not spatially disjoint. Treat these as within-region generalization estimates rather than strict geographic transfer performance.

Held-out Performance by Feature Set

target	feature_set	cv_r2_mean	cv_rmse_mean	stability_r2_mean	stability_rmse_mean	r2	rmse	mae	pearson_r
S1_VH	Embeddings only	0.931	1.238	0.934	1.246	0.942	1.182	0.858	0.971
S1_VH	Embeddings + region + Dynamic World	0.930	1.256	0.933	1.257	0.941	1.199	0.881	0.970
S1_VV	Embeddings only	0.903	1.155	0.907	1.149	0.902	1.255	0.825	0.951
S1_VV	Embeddings + region + Dynamic World	0.902	1.164	0.904	1.168	0.901	1.260	0.825	0.951
S1_VV_div_VH	Embeddings + region + Dynamic World	0.696	1.050	0.715	1.059	0.698	1.155	0.798	0.836
S1_VV_div_VH	Embeddings only	0.696	1.045	0.713	1.061	0.698	1.156	0.808	0.836

The holdout test set is shared across the two feature-set ablations. It is a stratified random holdout within sampled regions rather than a spatially disjoint holdout, so these estimates may be optimistic relative to stricter geographic transfer tests.

Repeated Grouped-CV Stability

target	feature_set	r2_mean	r2_std	rmse_mean	rmse_std	mae_mean	mae_std
S1_VH	Embeddings only	0.934	0.012	1.246	0.072	0.929	0.039
S1_VH	Embeddings + region + Dynamic World	0.933	0.010	1.257	0.077	0.946	0.050
S1_VV	Embeddings only	0.907	0.016	1.149	0.062	0.831	0.037
S1_VV	Embeddings + region + Dynamic World	0.904	0.016	1.168	0.070	0.846	0.043
S1_VV_div_VH	Embeddings only	0.713	0.071	1.061	0.051	0.793	0.026
S1_VV_div_VH	Embeddings + region + Dynamic World	0.715	0.069	1.059	0.060	0.795	0.041

Each row summarizes 12 grouped validation folds: 4 folds across 3 random seeds.

Regional S1_VV Diagnostics

region	feature_set	r2	rmse	mae	pearson_r	n_test
amazon_forest	Embeddings only	0.936	1.125	0.784	0.967	216
california_coast	Embeddings only	0.923	1.027	0.740	0.961	216
iowa_ag	Embeddings only	0.851	1.302	0.751	0.930	216
sf_bay_urban	Embeddings only	0.871	1.512	1.026	0.936	216
amazon_forest	Embeddings + region + Dynamic World	0.935	1.129	0.778	0.967	216
california_coast	Embeddings + region + Dynamic World	0.922	1.029	0.750	0.961	216
iowa_ag	Embeddings + region + Dynamic World	0.839	1.357	0.789	0.923	216
sf_bay_urban	Embeddings + region + Dynamic World	0.878	1.474	0.981	0.940	216

Region-level diagnostics on the held-out test set.

Land-use S1_VV Diagnostics

dw_label_name	feature_set	r2	rmse	mae	pearson_r	n_test
bare	Embeddings only	0.692	1.202	0.837	0.847	96
built	Embeddings only	0.741	2.070	1.227	0.876	96
crops	Embeddings only	0.769	1.006	0.772	0.878	96
flooded_vegetation	Embeddings only	0.930	0.744	0.528	0.965	96
grass	Embeddings only	0.863	0.960	0.697	0.931	96
shrub_and_scrub	Embeddings only	0.781	1.366	0.902	0.889	96
snow_and_ice	Embeddings only	0.897	1.020	0.737	0.948	96
trees	Embeddings only	0.861	0.894	0.715	0.930	96
water	Embeddings only	0.842	1.500	1.011	0.923	96
bare	Embeddings + region + Dynamic World	0.699	1.189	0.826	0.849	96
built	Embeddings + region + Dynamic World	0.732	2.108	1.233	0.872	96
crops	Embeddings + region + Dynamic World	0.782	0.978	0.724	0.885	96
flooded_vegetation	Embeddings + region + Dynamic World	0.917	0.810	0.559	0.958	96
grass	Embeddings + region + Dynamic World	0.858	0.978	0.691	0.929	96
shrub_and_scrub	Embeddings + region + Dynamic World	0.788	1.345	0.894	0.893	96
snow_and_ice	Embeddings + region + Dynamic World	0.898	1.016	0.761	0.948	96
trees	Embeddings + region + Dynamic World	0.847	0.939	0.745	0.923	96
water	Embeddings + region + Dynamic World	0.850	1.462	0.988	0.931	96

Land-use diagnostics on the held-out test set.

Feature Importance Summary

Top 5 features by permutation importance

Embeddings only

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S1_VV: A25 (0.120), A27 (0.089), A49 (0.063), A17 (0.039), A34 (0.030)
S1_VH: A25 (0.269), A27 (0.051), A49 (0.030), A19 (0.025), A34 (0.024)
S1_VV_div_VH: A25 (0.106), A24 (0.063), A39 (0.038), A63 (0.037), A30 (0.031)
```

Embeddings + region + Dynamic World

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S1_VV: water (0.113), A27 (0.084), A49 (0.055), A25 (0.041), A17 (0.037)
S1_VH: water (0.134), A25 (0.112), A27 (0.050), bare (0.031), A34 (0.025)
S1_VV_div_VH: A25 (0.066), A39 (0.049), A63 (0.041), A24 (0.041), A30 (0.027)
```
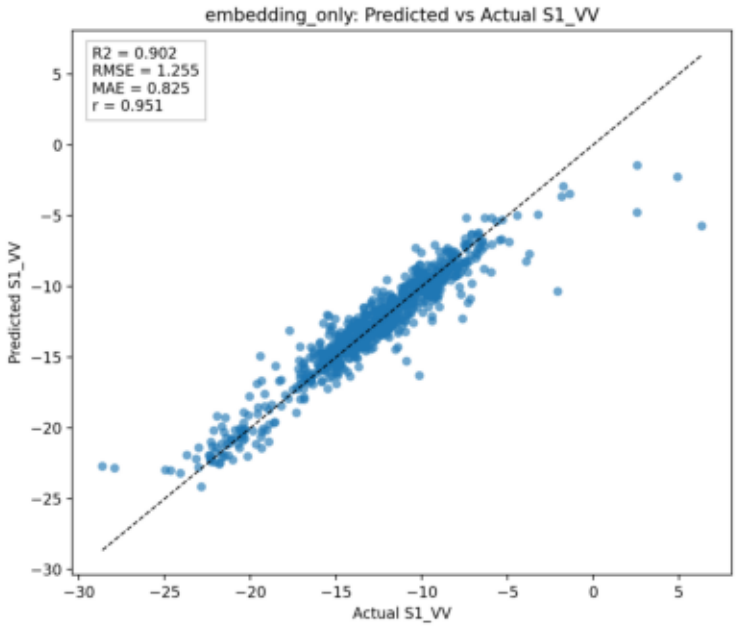
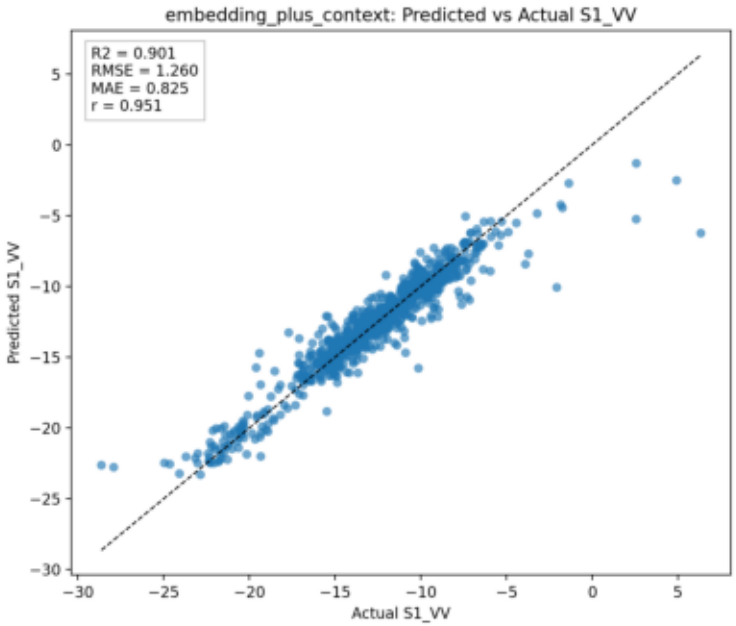

Predicted vs Actual

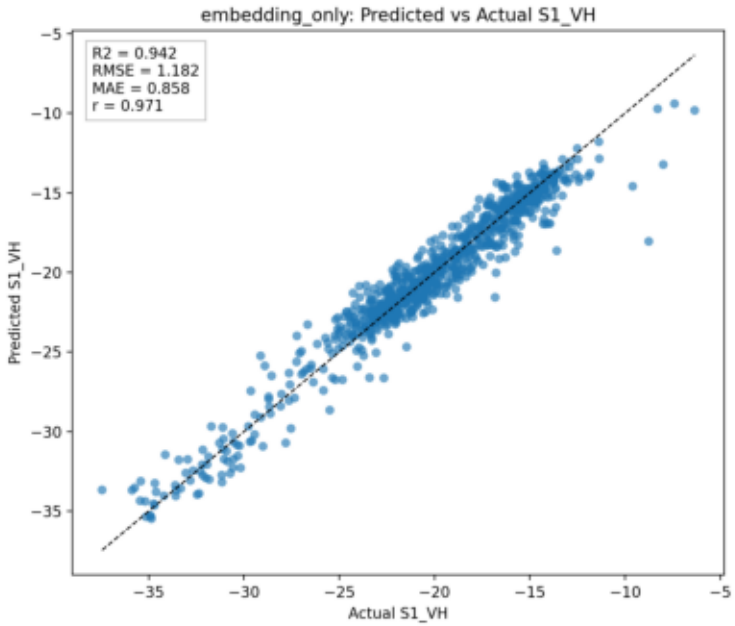
predicted vs actual embedding only S1 VV



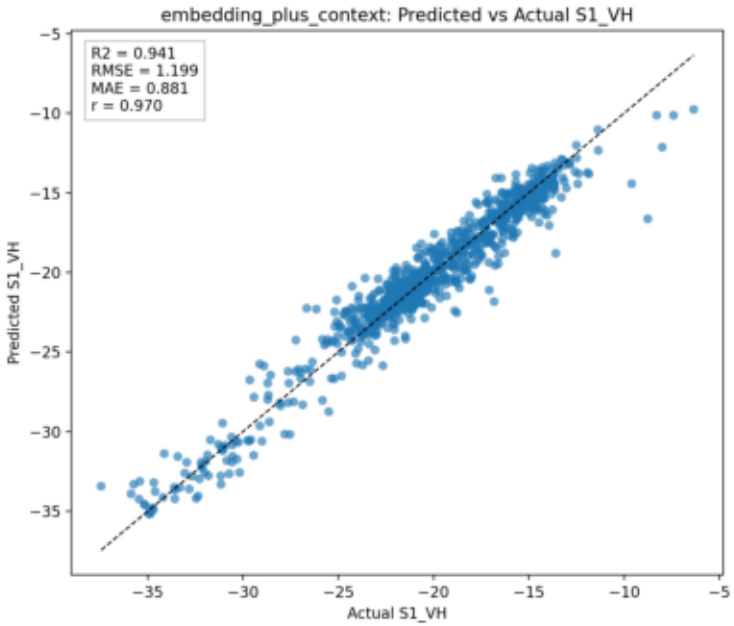
predicted vs actual embedding plus context S1 VV



predicted vs actual embedding only S1 VH

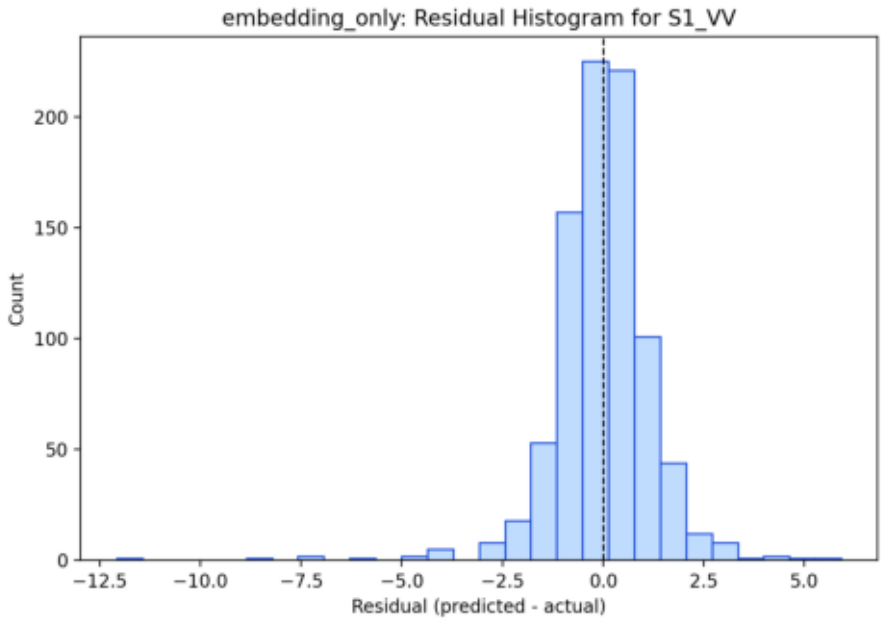


predicted vs actual embedding plus context S1 VH

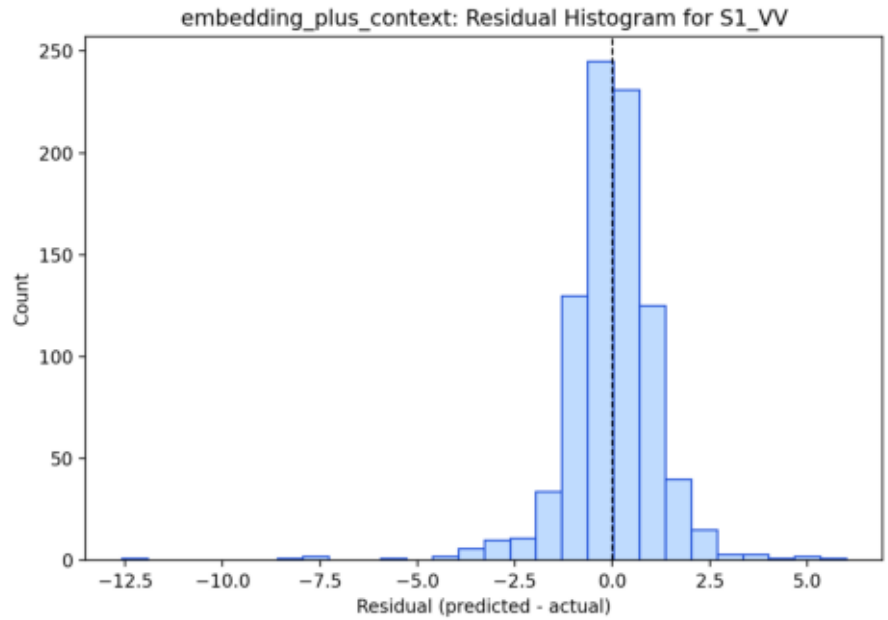


Residual Histograms

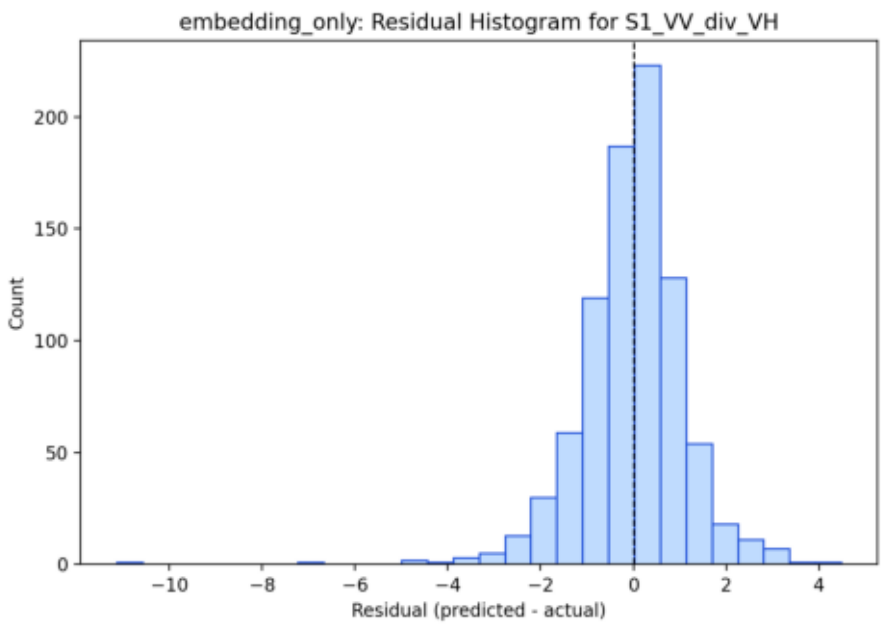
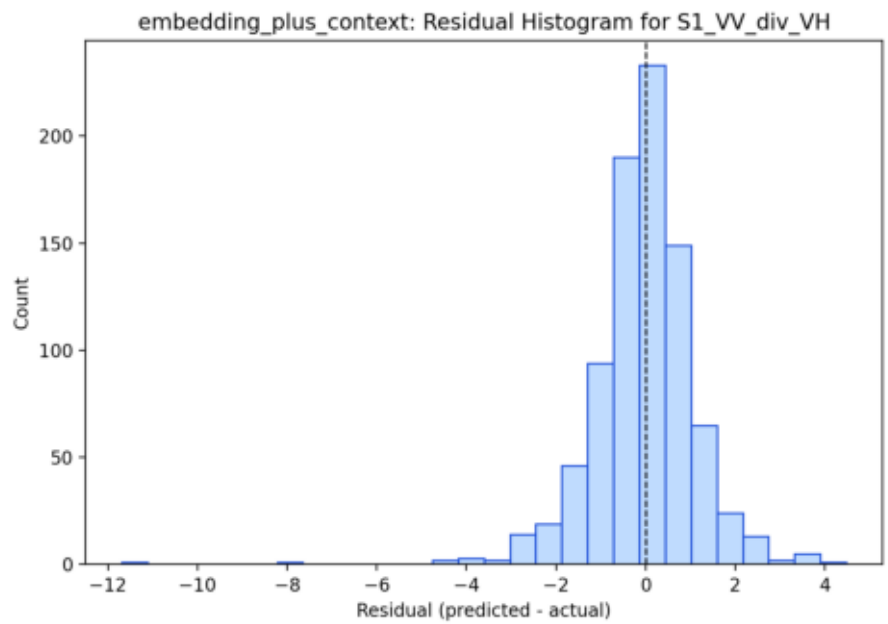
residual histogram embedding only S1 VV



residual histogram embedding plus context S1 VV



residual histogram embedding only S1 VV div VH



residual histogram embedding plus context S1 VV div VH